AYAM BANJADE

Github Linkedin Portfolio

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Bremen, Germany

PROFILE

Motivated and disciplined CS student passionate about technology and software development. As a fast learner, I am eager to tackle complex problems and adopt new tools and technologies.

EDUCATION

B.Sc. in Computer Science — minor in Mathematics
(Sept. 2021 - Jun. 2024)

Constructor University Bremen

Relevant Courses:

- Software Engineering
- Databases and Web Services
- Algorithms and Data Structures
- Computer Networks
- Operating Systems
- Distributed Algorithms
- Secure and Dependable Systems
- Machine Learning
- Artificial Intelligence

SKILLS

- C/C++, Rust, Java, Python, PHP, JavaScript, TypeScript, Flutter, Kotlin
- Svelte, jQuery, React, Next.js, HTML, CSS, Tailwind, npm
- MySQL, MongoDB, XML, JSON
- Linux, Shell Script, CLI, Git, Docker

LANGUAGES

English: Native/Bilingual (C2)
German: Beginner (A2.2)

EXPERIENCE

Teaching Assistant at Constructor University Bremen

Numerical Methods — Feb. 2023 - Jun. 2023

- Created and solved challenging problems and helped 100+ students with numerical methods for root finding, interpolation, approximation, numerical differentiation / integration / differential equations, and their implementations in Python
- Streamlined grading process by developing Python scripts to transform and simplify complex CSV files

Programming in C/C++ — Sept. 2022 - Dec. 2022

- Provided comprehensive tutorials to 200+ students on programming concepts, including pointers, object-oriented programming, dynamic memory allocation, data structures, templates, macros, header files, and makefiles
- Debugged and validated C/C++ code, ensuring students' understanding and proficiency

Introduction to Computer Science — Sept. 2022 - Dec. 2022

- Coordinated and led tutorials addressing 95% of all CS students' inquiries in functional programming (Haskell), assembly programming, computer science foundations, discrete mathematics, Boolean algebra, Hoare logic, and digital circuit design

LEADERSHIP ACTIVITIES

Member of CS Hiring Commitee — **Jun. 2023 - Present**
Constructor University Bremen

- Serving as the representative for the Computer Science student body in the hiring process for faculty, actively casting votes and advocating for the inclusion of student interests

Member of Makers and Coders — **Aug. 2019 - May 2021**
Ullens School (IBDP / High School)

- Coordinated seminars with STEM experts, conducted workshops for underprivileged Nepali students, and managed fundraising events for required equipment for schools and the club

PROJECTS

Rippit

- Developed a frontend and PHP backend of a forum website collaboratively using MySQL, MongoDB, jQuery, and VCS (git)
- Implemented post types, post rating, post sorting, user authentication, tickets (feedback, reports), and search suggestions, enhancing user engagement and content creation and discovery
- Utilized data definition languages to implement the entity-relationship model and query-dependent access rights using sanitized SQL queries for enhanced security and efficient data retrieval and analysis
- Designed homepage, individual post page, user profiles page, and admin users management

Bug World

- Collaborated in agile sprints to build and improve different code bases for a full-stack simulation game in JavaScript and HTML/CSS
- Engineered and implemented a parser, assembler, simulator, and GUI using OOP principles
- Created automated test cases (Jest, Jasmine) to ensure robustness and functionality
- Authored and maintained detailed technical documentation and comprehensive specifications to ensure effective project communication and development

Room Booking System

- Designed and built a Java-based, user-friendly, and intuitive CLI room booking system with file I/O, dynamic matching, and custom data structures for efficient data management

Spam Email Extractor

- Built a C++ program to implement data extraction techniques to retrieve 57 relevant attributes from UCI's Spambase Data Set for effective spam email detection

Stochastic Methods Lab

- Developed stochastic models for financial mathematics using Python libraries (matplotlib, NumPy, pandas, SciPy) for CSV handling and complex dataset analysis and visualization

Bremen, Germany, 2023-07-02 Ayam